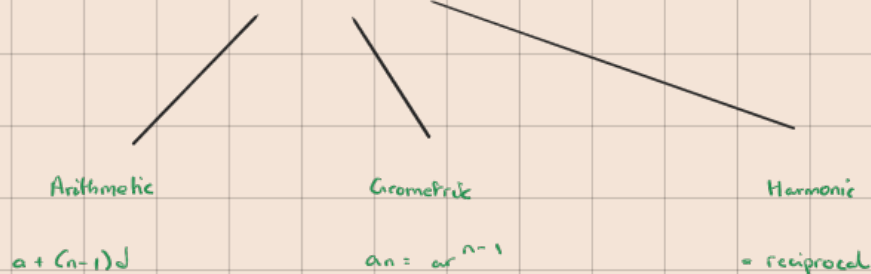
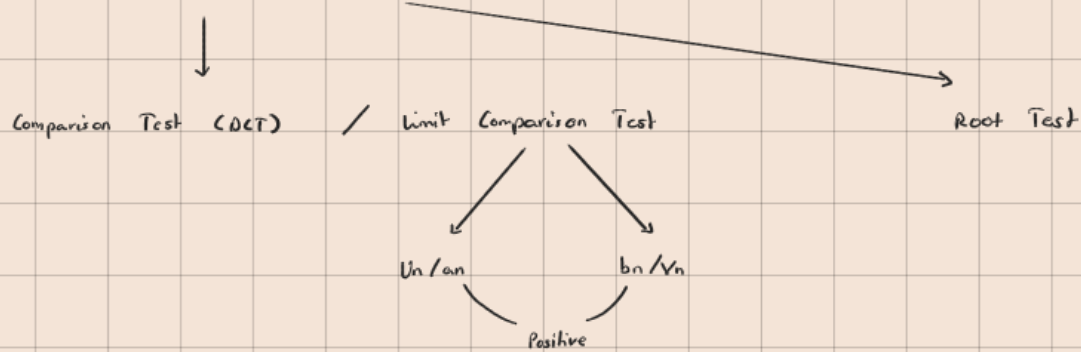


wed, Feb 18

Sequence and Series



Application of Convergence



$$\frac{a_n}{b_n} \lim_{n \rightarrow \infty} = \underline{\quad ? \quad} \leftarrow \text{Defined}$$

(Helping (P-Series) Test)

$$b_n \text{ or } v_n = \underline{\quad 1 \quad}$$

Something

$$= \frac{1}{n^p} \rightarrow \begin{array}{l} p > 1 \text{ converge} \\ p \leq 1 \text{ Diverge} \end{array}$$

$$\textcircled{a)} \quad \sum_{k=1}^{\infty} \frac{3k^3 - 2k^2 + 4}{k^7 - k^3 + 2}$$

$$U_n = \frac{3k^3 - 2k^2 + 4}{k^7 - k^3 + 2}$$

$$V_n = \frac{3k^3}{k^7} = \frac{3}{k^4} \text{ or } \frac{1}{k^4}$$

p-series

$$\frac{1}{k^4} \text{ or } \frac{3}{k^4}$$

$$\lim_{n \rightarrow \infty} \frac{3k^3 - 2k^2 + 4}{k^7 - k^3 + 2} \times k^4$$

$4 > 1 \approx \text{converge}$

$$= \frac{3k^7 - 2k^6 + 4k^4}{k^7 - k^3 + 2}$$

$$= \frac{k^4(3)}{k^7(1)} \approx 3$$

$$c) \quad \frac{5}{4} + \frac{7}{9} + \frac{9}{16} + \dots$$

$$U_n = \sum_{n=1}^{\infty} \frac{2n+3}{(n+1)^2}$$

$$\frac{2n+3}{n^2+2n+1} \approx \frac{n}{n^2} = \frac{1}{n}$$

$$\frac{2n+3}{n^2+2n+1} \times n$$

$$\frac{\cancel{2n} + 3n}{n^3 + \cancel{2n^2} + n}$$

$$= \frac{3}{n^2} = \frac{1}{n}$$

$$d) \quad \sum_{n=1}^{\infty} \frac{2n^2 + 3n}{5 + n^5}$$

$$\approx \frac{n^2}{n^5} = \frac{1}{n^3}$$

$$\frac{2n^2 + 3n}{5 + n^5} \times n^3$$

$$= \frac{\cancel{2n^2} + 3n^4}{5 + \cancel{n^5}} = \frac{2n^5 + 3n^4}{5 + n^5}$$

$$= \frac{\cancel{n^4}}{\cancel{n^4}} = \frac{n^5 (2 + \frac{3}{n})}{n^5 (\frac{5}{n^5} + 1)}$$

$$= n^4$$

$$= \frac{2}{1}$$

$$e) \quad \sum_{n=1}^{\infty} \frac{1}{n^2+1}$$

$$\sum_{n=1}^{\infty} \frac{O(1)}{n^2(1+1/n^2)} = 0$$

